MESOPOTAMIA · A MODERN AUDIBLE REALIZATION

Sammû · listening practice

2 retuning moves · initial string 1: 440 Hz · companion strings: octave · sound: sine

An audible fourths-and-fifths realization, not a recovered ancient performance. Six designated pairs are set to frequency ratios 4/3 or 3/2. Reference pitch, numerical precision, sound colour and the companion-string octave/unison choice are modern settings. The tablets do not record frequencies in hertz or these fractions.

Your tentative adjustment

The two selected partials

Open dot: unchanged frequency. Filled dot: your setting. Both rows share a fixed linear axis, from 983.7 to 1260.5 Hz.

String 6 · partial 4 held fixed		1042.962963 Hz
String 3 · partial 3 adjusted		1045.185319 Hz

Frequency difference: **2.222356 Hz**

When these two partials are isolated as steady sine tones, their frequency difference is the amplitude-beat rate. In the full synthetic pair it is one interaction among several partials.

Your setting and the model's answer

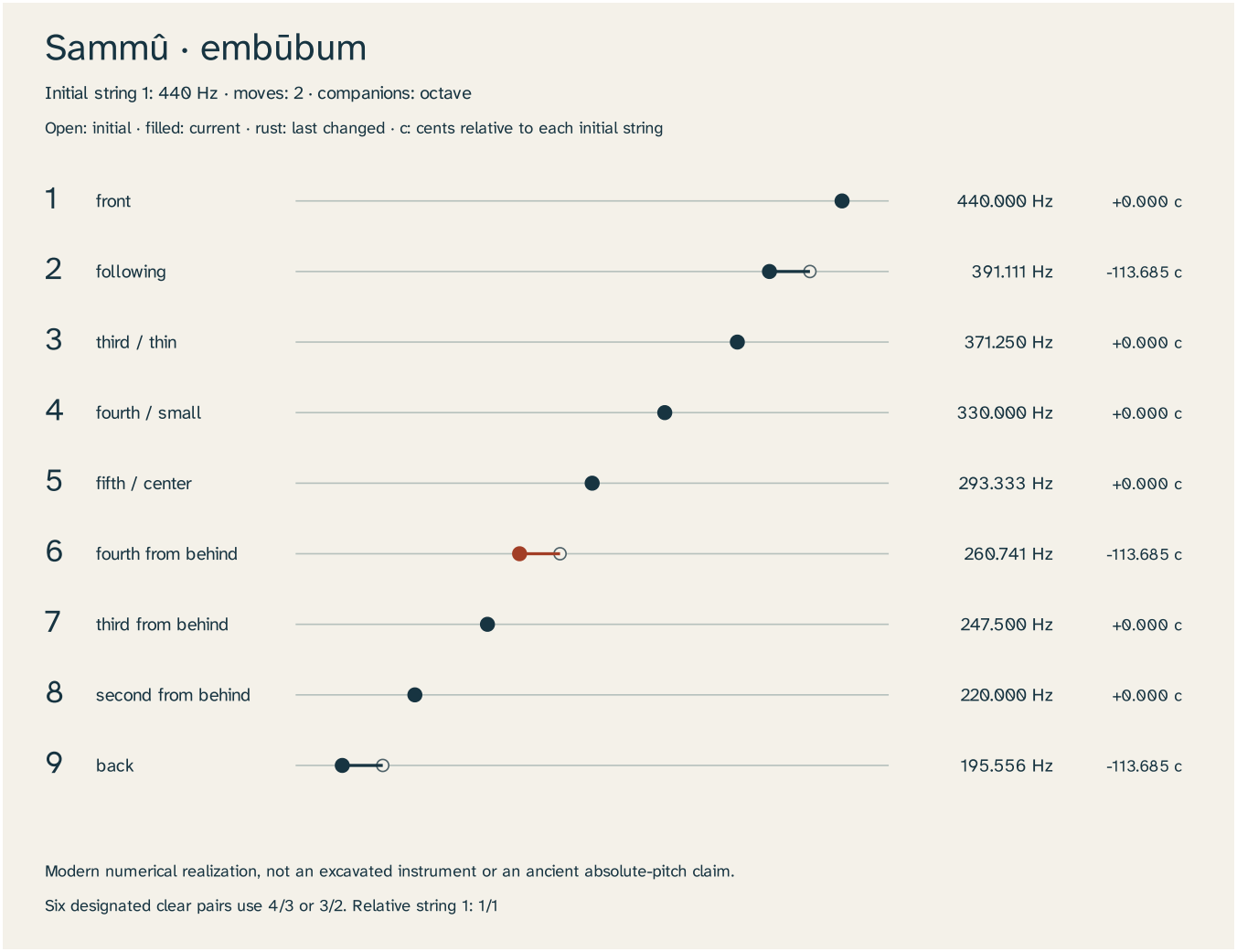
Your loosening	110.000000 cents
Model's exact adjustment	113.685006 cents · × 2048/2187
Difference from the target	-3.685006 cents

Negative means short of the target; positive means beyond it. This is a numerical comparison, not a pass/fail judgment about your hearing or a threshold for the ancient term *zakû*.

The underlying tuning remains embūbum. The model's exact operation changes string 3 and reaches pītum; your tentative setting is kept separately.

Reopen this listening practice

Underlying exact tuning, before the preview



Hear the experiment

Unchanged, then your adjustment: strings 6 + 3, isolated partials. Each sound is synthetic. The two selected partials are isolated as equally loud, steady sine tones.

Current pitches

Exact ratios are relative to the initial first string, not a new reference after every move. Shifts compare each string with its own initial pitch.

String	Ratio to initial 1	Hertz	Shift ratio	Shift, cents
1	1/1	440.000000	1/1	0.000000
2	8/9	391.111111	2048/2187	-113.685006
3	27/32	371.250000	1/1	0.000000

String	Ratio to initial 1	Hertz	Shift ratio	Shift, cents
4	3/4	330.000000	1/1	0.000000
5	2/3	293.333333	1/1	0.000000
6	16/27	260.740741	2048/2187	-113.685006
7	9/16	247.500000	1/1	0.000000
8	1/2	220.000000	1/1	0.000000
9	4/9	195.555556	2048/2187	-113.685006

Every retuning step

Step	Action	Resulting tuning	Strings changed	String 1 ratio
0	initial	išartum	—	1/1
1	loosen	kitmum	2 + 9	1/1
2	loosen	embūbum	6	1/1

Read the procedure and its preservation

These are our summaries of three neighboring cases in the published translation. A supplied scribal omission and a restoration across damaged text are different kinds of editorial intervention.

išartum → kitmum

Begin in išartum. Test qablītum, strings 5 + 2. Loosen the following string and the back string: 2 + 9. The resulting tuning is kitmum.

The word describing the pair as unclear is supplied by the editor as an omission, marked ⟨ ⟩ in the edition. The case is not an entirely untouched ancient sentence.

Loosening section, lines 13–16; Mirelman 2013, pp. 49–52.

kitmum → embūbum

Begin in kitmum. Test išartum, strings 2 + 6. Loosen the fourth string from behind: 6. The reconstructed result is embūbum.

The result clause is restored across a break, marked [ ] in the edition. The diagnostic pair and action are preserved in the published translation.

Loosening section, lines 17–20; Mirelman 2013, pp. 51–52.

embūbum → pītum

The reconstructed case tests kitmum, strings 6 + 3, and loosens string 3. Its destination is pītum.

Almost all of this case is restored in [ ]; the final tuning name survives. Its place in the pattern is an editorial reconstruction, not an equally preserved instruction.

Loosening section, lines 21–24; Mirelman 2013, p. 52.

Akkadian word notes

Normalized romanization, with selected meanings in this context. The notes distinguish a preserved action word from a restored interpretation.

sammû	**a stringed instrument** Its precise identification as harp or lyre is unsettled. The diagram here describes string relationships, not its archaeological shape.
zakû	**clear / clean / pure** A source term for a pair of strings. Its musical meaning is interpreted; it is not a recorded numerical frequency ratio.
lā zakû	**not clear** The diagnostic pair distinguishes one named tuning from another. The spelling here is normalized Akkadian romanization.
išartum	**straight / normal** Both the name of strings 2 + 6 and the name of a tuning. Pair names and tuning names do different jobs.

kitmum	**closed** A tuning name and a pair name: strings 6 + 3. The same pair can be clear in one tuning and unclear in another.
qablītum	**middle** Names strings 5 + 2. In išartum tuning this is the diagnostic unclear pair; after loosening string 2 it is clear.
nê'um	**to loosen** Intact and partly preserved forms occur in the loosening chapter. They support the physical direction assigned to that action. See Rahn (2022), paragraph 45.
nasāḫum	**to remove / subtract; ‘tighten’ here** Editors restore the verb in the tightening instructions. A partly preserved concluding rubric supports this contextual interpretation. It does not mean ‘tighten’ in every Akkadian setting. See Rahn (2022), paragraph 45.
šumma	**if** Introduces a condition. In a procedural case, distinguish that condition from the instructed action and its result. Lexical support: Høyrup (2017), Appendix B; the tuning text’s conditional format is discussed by Mirelman (2013), pp. 49–50.

How to read this model

Six pairs of strings are assigned exact 4/3 or 3/2 frequency ratios. The remaining diagnostic pair is left unclear. Each operation changes only the prescribed string (and its companion for strings 1 or 2), making that diagnostic pair clear. The next diagnostic pair then becomes unclear.

Seven moves in one direction return the tuning name but shift every string by 2048/2187 when loosening, or 2187/2048 when tightening. These fractions are consequences of this model, not ratios written on a tablet. No ancient reference pitch, recorded performance, instrument shape, or exact ancient sound colour is claimed.

The two tuning manuscripts from Ur are fragmentary. Combining them with later string-pair evidence is a scholarly reconstruction. In Mirelman’s edition, supplied scribal omissions use ⟨ ⟩ and text restored across a break uses [ ]. For example, the first loosening case has an editorially supplied description of the diagnostic pair; the third is mostly restored.

Sources and provenance

1. Sam Mirelman · Tuning Procedures in Ancient Iraq (2013)

   Analytical Approaches to World Music 2(2), 43–56. String names: p. 46; two manuscripts and uncertain instrument: pp. 47–48; preserved/restored clauses: pp. 49–52; seven tunings: fig. 5, p. 52; tightening: pp. 53–54.

2. Jay Rahn · Was Mesopotamian Tuning Diatonic? A Parsimonious Answer (2022)

   Music Theory Online 28.1. DOI 10.30535/mto.28.1.7. Paragraphs 9–16 distinguish physical strings from later pitch concepts and explain the reversible seven-name network; later sections examine assumptions behind an audible reconstruction.

3. Sam Mirelman & Theo J. H. Krispijn · The Old Babylonian tuning text UET VI/3 899 (2009)

   Iraq 71, 43–52. Edition of the second manuscript. Read with UET VII 74; the two witnesses are separate tablets, not joining fragments of one tablet.

4. CDLI P347039 · UET VII 74

   Catalogue record for one tuning manuscript from Ur, Old Babylonian period. The British Museum witness discussed by Mirelman is a cast; do not infer the original tablet's present location from the catalogue's collection field.

5. CDLI P254475 · CBS 10996

   Later witness for named string pairs. The catalogue checked on 22 October 2026 assigns Middle Babylonian, approximately 1400–1100 BCE; Mirelman (2013) used an older Neo-Babylonian dating. Combining witnesses is a scholarly reconstruction across time.

6. Jens Høyrup · Algebra in Cuneiform, Appendix B (2017)

   Key to Vocabulary and Standard Translations: šumma, 'if'. This is lexical support from a study of mathematical texts, not another witness to the musical procedure. The action verbs are discussed by Rahn (2022), paragraph 45.

Reopen the experiment in Sammû

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